GeoCORK: User Manual

Geologic Cataloger, Organizer, Record Keeper

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Abstract

This user manual for GeoCORK aims to guide new users through the complete functionality of our research data management tool for geochronological data—from opening or creating databases, importing, filtering, editing, and exporting geologic and geochronologic data.

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1 Introduction

GeoCORK's intended use is purposefully left vague as different geoscience disciplines will have different use-cases for GeoCORK. Therefore, GeoCORK's design of tables, attributes, and options listed are left largely to the interpretation of the researcher and their specific needs. Through this user manual researchers unfamiliar with GeoCORK will learn tips, tricks, and how to efficiently utilize GeoCORK to it's fullest.

Main Widgets GeoCORK has a few features we call the main widgets. These main widgets include the `Data Viewer Widget`, `Filters Tab`, `Export Tab`, `Import Wizard`, `Filtered Data View`, `Data View Tab`, `Edit Table Dialog`, `Add Item Dialog`, and `Sample Edit Dialog`.

Data Viewer The `Data Viewer Widget` is the primary way to view all data within a database. A combo switcher allows quick changing between tables in the database. Tables are paged and can be sorted by clicking column headers. Trees are not paged, and item order is stored in the database.

Filters The `Filters Tab` allows complex multivariate and nested logical operators to filter all data within the database. Filtered results can be shown as Samples, Aliquots, Grains, Spots, or U-Pb Analyses.

Export The `Export Tab` allows any data, selected manually or through the Filter widget, to be saved as an Excel, CSV, or database file in pre-defined or custom formats.

Import The `Import Wizard` allows data in an Excel (.xlsx only) file to be imported into GeoCORK. Multiple tabs are supported. Assisted importing through column selection, row removal/rejection, and metadata selection enables quick importing. Import mappings can be saved for future one-click use.

Filtered Data View The `Filtered Data View` provides a high-level overview of Samples, Aliquots, Grains, Spots, or U-Pb Analyses metadata across the entire database which pass the filter in the `Filters Tab`.

Data View Tab The `Data View Tab` creates a tab view of the Aliquots, Grains, Spots, or U-Pb Analyses of any parent Sample, Aliquot, Grain, or Spot. It is impractical to view all analyses in the database at once, so they can only be viewed by right clicking on a parent item.

Edit Table Dialog The `Edit Table Dialog` allows editing fields in the table by double clicking a cell or selecting multiple cells in a column and right clicking. Some fields (e.g. GPS) bring up an additional dialog window to ensure data integrity across multiple fields. Tree items can be moved by drag-drop.

Add Item Dialog The `Add Item Dialog` allows the user to add new items to a table. In trees, items can be added as a parent, child, before a sibling, after a sibling, or at the end.

Sample Edit Dialog The `Sample Edit Dialog` allows editing metadata for one or more samples. This view is the primary way to modify sample metadata through text boxes, drop-down menus, or check boxes.

Tables All of GeoCORK's tables are listed below. GeoCORK requires items in each table to be unique across all other names within the same table. This is to ensure that when using GeoCORK's other functionality, especially within

filters, the user will not be presented with two options that have the exact same name (e. g.,Basalt and Basalt), causing confusion as to which one to select. GeoCORK has 59 tables in total to accomplish all functionality. Most of the tables within GeoCORK are not user viewable and used internally. For completeness all table and their purpose are listed below.

1.1 Core Sample-Aliquot-Grain-Spot-U-Pb Tables

Samples	Primary table storing sample-level data, names, metadata, and relationships.
Aliquots	Aliquot-level records associated with samples.
Grains	Grain-level entries.
Spots	Spot-level records associated with aliquots and optionally with grains.
UPbAnalyses	U–Pb analysis records, with all ratios, errors, and metadata associated with spots.

1.2 Core Context/Composition Tables

The context and composition tables used in the Samples, Aliquots, Grains, Spots, and UPbAnalyses tables are intended to store metadata that may not fit in other metadata tables. Since, like all other tables, metadata in these tables can be associated with many samples, and one sample be associated with many contexts or a single composition. The context tables could be used for adding categorical variables at any level in the Sample to U-Pb Analysis hierarchy.

The composition tables could store mineral information for a given grain or spot. Our idea behind separating these two instead of a combined approach is that Contexts should be used for broad categories that may be used across many sample metadata, while compositions could be used for grain or spot specific quirks or notes. An example of contexts may be to store the analyst who ran the instrument while analyzing a specific spot. The composition table could store geologically relevant metadata, such as the specific mineral or inclusions seen. As with all of GeoCORK these examples are not the only way to utilize the tables and each researcher is encouraged to utilize them best for their own individual workflows and projects.

Sample Contexts	Context categories for samples (e.g., outcrop, core, block, float).
Aliquot Contexts	Classification or contextual tags for aliquots.
Grain Contexts	Context labels for grains (e.g., morphology, alteration features).
Grain Compositions	Stores mineral or chemical composition metadata at the grain level.
Spot Contexts	Defines contextual metadata about a spot (e.g., rim/core, zoning, inclusion, morphology).
Spot Compositions	Stores compositional metadata tied to individual spots.
U-Pb Analysis Contexts	Stores context categories for U–Pb analyses (e.g., analysis conditions, depth of spot).

1.3 Metadata Tables

The metadata tables are the primary way to organize sample, geological, age, or contextual metadata within GeoCORK. All tables are purposefully left vague or with little guidance on how best to incorporate or populate with data.

1.3.1 Core Geologic Metadata

GPS Locations Linked with the Samples and Columns tables. Stores GPS coordinate data and metadata in various common formats (i. e.,DD, DDM, DMS, standard UTM).

Sampling Methods Linked with the Samples table. Allowed sampling method descriptors (e.g., hand sample, drill core).

Rock Types Linked with the Samples table. Rock Types may refer to typical lithological nomenclature or mineral assemblages.

Units Linked with the Samples table. Units may refer to mapped geological units, biostratigraphic formations, lithostratigraphic formations, or economic prospectivity zones.

Settings Linked with the Samples table. Settings may refer to broad tectonic, depositional, or emplacement models. They may also refer to regionally specific settings.

Regions Linked with the Samples table. Regions may refer to broad geographic tags such as countries, counties, or cities. They may also refer to watersheds, geographic features, or field areas.

Columns Linked with the Samples table. Represents stratigraphic columns, wells, or cores that samples can be attached to.

1.3.2 Geologic Ages

Ages Ages table contains the GSA Timescale v6.0 for all Periods, Epochs, and Ages.

Sample Ages Linked with the Samples table. Sample Ages have fields for direct and relative ages.

Age Constraints Linked with the Sample Ages table. Enables a Sample Age entry to have a descriptor of how the age is constrained (e. g.,MDA method, biostratigraphy).

Age Interpretations Linked with the Samples Ages table. Enables a Sample Age entry to have an interpretation to describe the interpreted meaning of an age (e. g.,depositional, metamorphic).

Age Signatures Linked with the Samples table. Descriptive signatures or fingerprints for U-Pb age populations present in a sample.

1.3.3 Analytical Metadata

Instruments Linked with the U-Pb Analyses table. Lists all analytical instruments referenced in U–Pb or geochemical analyses.

Lab Facilities	Linked with the U-Pb Analyses table. Stores laboratories or analytical facilities where analyses were performed.
U-Pb Analysis Methods	Linked with the U-Pb Analyses table. Lists available U–Pb analytical methods (e.g., LA–ICP–MS, ID–TIMS).
Rejection Reasons	Linked with the U-Pb Analyses table. Defines allowed reasons for marking U–Pb analyses as rejected.
References	Linked with the U-Pb Analyses and Sample Ages table. Contains bibliographic references with fields for authors, title, year, journal, and DOI.

1.4 Internal Format/Error Tables

AgeUnits	Stores allowed units for age values (e.g., Ma, ka, years).
ConcordanceFormats	Defines supported concordia / concordance reporting formats for U–Pb analyses (e. g.,concordance, discordance, percent, absolute).
DirectionUnits	Stores north, south, east, and west for reference in GPS fields.
DistanceUnits	Stores units used for length, distance, height, or depth fields.
ErrorFormats	Defines acceptable formats for analytical uncertainties (absolute, percent, 1 or 2σ, etc.).
GPSFormats	Lists supported GPS coordinate input and display formats.
FilterGroups	Stores user-defined filter names and associated queries.
AgeUnitConversions	Stores conversion calculations between different age units.
ConcordanceFormatConversions	Stores conversion calculations between differing concordance reporting formats.
DistanceUnitConversions	Stores conversion calculations between distance and depth units.
ErrorFormatConversions	Stores conversion calculations between error reporting styles.
GPSFormatConversions	Stores conversion calculations between GPS coordinate formats (DD, DMS, etc.).

1.5 Linking Tables

	These are many-to-many relationships like tags. A sample can have multiple tags, and a tag can be associated with multiple samples.
SampleAges_References	Associates sample ages with bibliographic references.
SampleAges_AgeConstraints	Associates sample ages with constraint types.
SampleAges_AgeInterpretations	Associates sample ages with interpretations.
Samples_SampleAges	Associates samples with sample ages.
Samples_SampleContexts	Associates samples with sampling contexts.
Samples_SamplingMethods	Associates samples with sampling methods.
Samples_RockTypes	Associates samples with lithologic rock type classifications.
Samples_Units	Associates samples with geologic units.
Samples_Regions	Associates samples with geographic/tectonic regions.

Samples_Settings	Associates samples with user-defined settings.
Samples_AgeSignatures	Associates samples with U-Pb age signatures.
Aliquots_AliquotContexts	Associates aliquots with contextual labels.
Grains_GrainContexts	Associates grains with contextual labels.
Spots_SpotContexts	Associates spots with contextual labels.
UPbAnalyses_UPbAnalysisContexts	Associates U–Pb analyses with contextual labels.
UPbAnalyses_RejectionReasons	Associates rejected U–Pb analyses with rejection reasons.

2 Getting Started

2.1 Landing Page



When GeoCORK launches the first window seen is called the *Landing Page*. This page provides the only entry into the rest of GeoCORK. Three buttons are present on this page: New Database, Open Database, and Merge Database. The citation at the bottom is a link that will open GeoCORK's DOI and the GitHub logo will bring you to our GitHub page.

Open Existing Database: Click **Open** to open a file selector prompt. Navigate to the desired *.db* file and open. A list of previously opened databases is seen on the left.

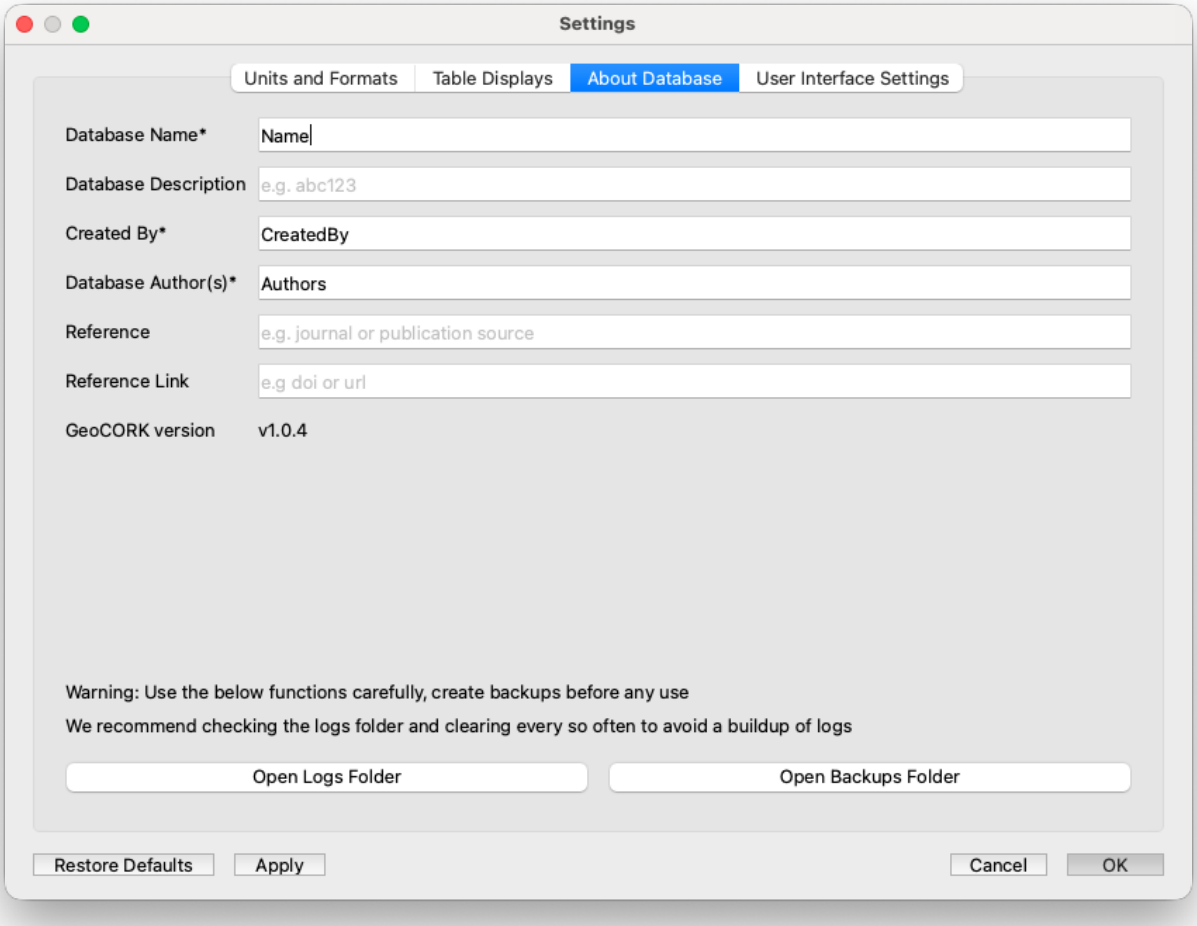


Figure 1: About database dialog when creating a database.

An item in this list may be double-clicked to quickly open a database or single-clicked/highlighted and then pressing **Open** to proceed.

Create New Database: Click **New** to open a file creator prompt. Navigate to the desired *.db* file location and provide a filename.

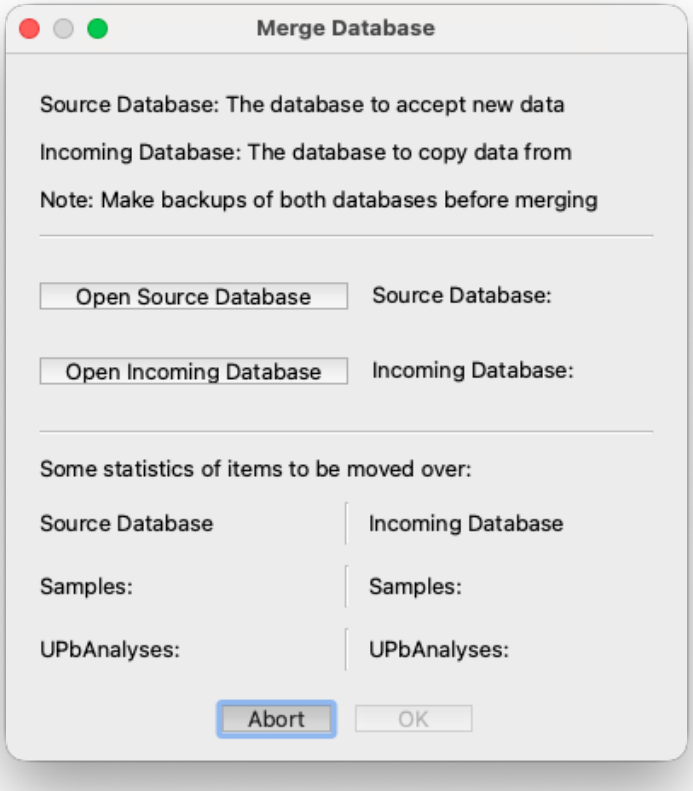
After creation, an optional metadata popup will be prompted to enter the following:

- Database Name
- Database Description
- Created By
- Database Author(s)
- Reference
- Reference Link

All of these settings can be modified later in the *Settings/Preference* window under **File** in the top toolbar of the application.

Merge Two GeoCORK Databases:

Before merging we strongly encourage to backup both databases using the `File → Create backup` in the main toolbar. The **Source Database** receives the merged data, while the **Incoming Database** contains the data to be added/merged over. During merging, there may be conflicts of duplicated names. To overcome this, GeoCORK will append "(1)" and "(2)" to resolve. We recommend looking over and cleaning the appended data in GeoCORK.



Summary statistics will be shown at the bottom indicating how many Samples and U-Pb data will be merged.

2.2 User Settings / Preferences

The user settings or preferences page is accessed via the toolbar or CMD+,. This page contains all settings or configurations for display settings that are persistent across all of GeoCORK, no matter which database file is open. Therefore two different databases will "share" the same display settings. Click `Apply` to save, `OK` to save and close, `Cancel` to discard any changes and close, or `Restore Defaults` to return all settings to default GeoCORK

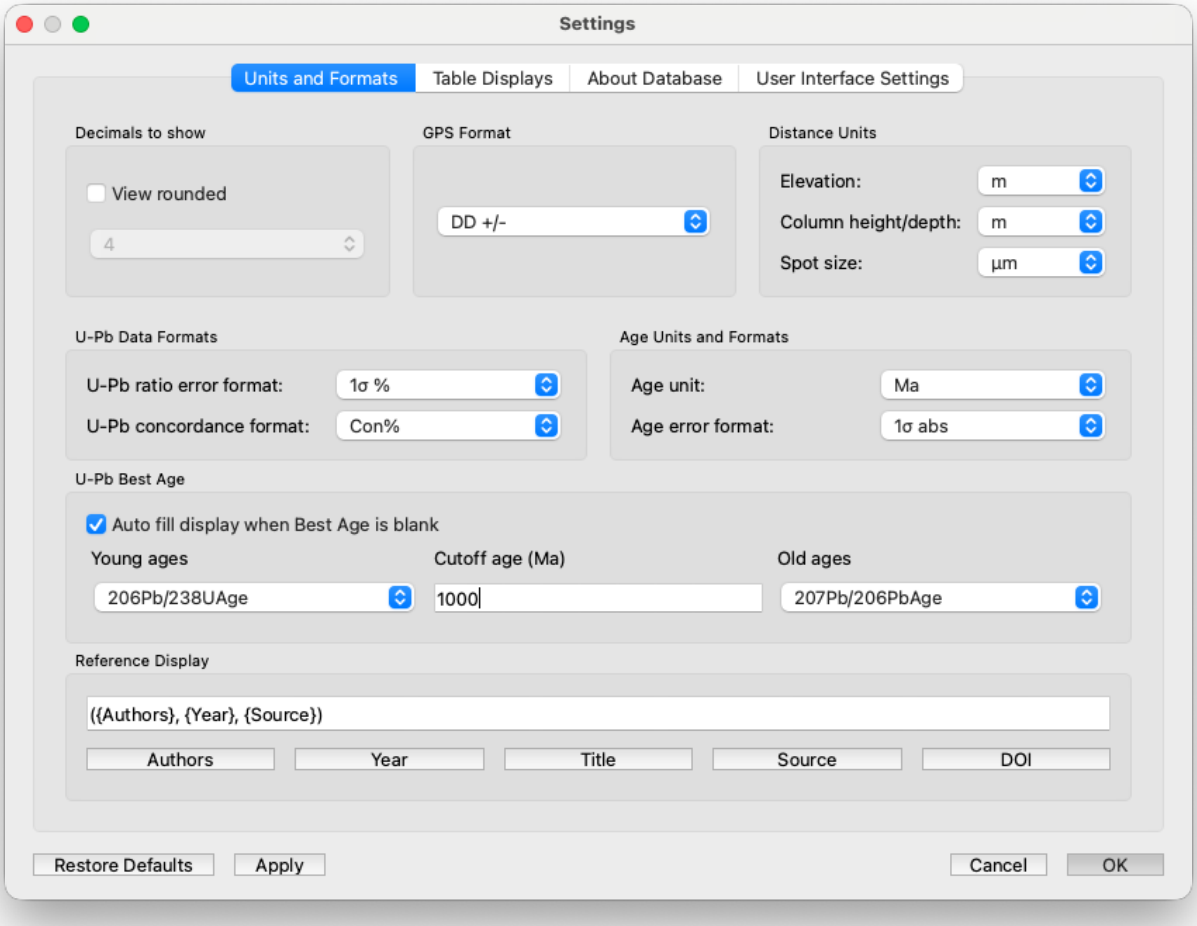


Figure 2: Units and formats tab in the settings window.

settings.

2.2.1 Units and Formats

Decimals To Show A checkbox indicates an enable/display to round decimals and a dropdown allows the amount of decimals to show.

GPS Format A dropdown to switch between any of the available GPS formats. This will not modify any stored GPS data. All tables or displays with GPS data are merely converted on the fly to display this unified format.

Distance Units There are three dropdowns for the displayed distance units. Elevation is used for GPS metadata. Column height/depth is used in the Samples table. Spot size denotes the diameter size of a laser-ablated hole on a grain.

U-Pb Data Formats There are two dropdowns for U-Pb geochronology data. The U-Pb ratio error format denotes the error for isotope ratios (e. g., $^{207}Pb/^{235}U$ with

$\pm 2\sigma$ %) and has four available formats: 1σ absolute, 2σ absolute, 1σ percent, 2σ percent. The U-Pb concordance format has four available formats: Concordance, Concordance Percent, Discordance, and Discordance Percent.

Age Units and Formats: There are two drop downs for age data. The age unit is the displayed value for age data, either in Samples, SampleAges, or U-Pb data. The age error format has four available formats: 1σ absolute, 2σ absolute, 1σ percent, 2σ percent.

U-Pb Best Age: The U-Pb Age allows auto filling of blank best age values for displaying and exporting. Automatic fill does not override the blank values present in the database, only when displaying or exporting. Three data entries are present: Young age value, cutoff age (Ma), and old ages. As shown the Best Age will be the $^{206}Pb/^{238}U$ age when the $^{206}Pb/^{238}U$ age is less than (<) 1000 Ma and the $^{207}Pb/^{208}Pb$ age otherwise.
Available values for ages are: $^{206}Pb/^{238}U$, $^{207}Pb/^{235}U$, $^{207}Pb/^{206}Pb$, $^{208}Pb/^{232}Th$

Reference Display: The reference display is a combined line edit with parsed variables. The Authors, Year, Source, DOI represent the attributes present in the References table. Clicking these buttons will auto populate the line edit. Other text can be inputted but will be present in all reference displays. Changing this field will update the reference display. No data are permanently changed within the database.

2.2.2 Table Displays

The table displays tab contains the settings for how all tables display to the researcher. Font size, scalers, and column selections are available.

Show per page: A global dropdown to modify how many records/rows are displayed in each page of a table. Available options are: 10, 25, 50, 100, 250, 500, and 1000. Tables will automatically display with the selected number of rows, but this can be manually adjusted in the table view as well.

Show items with missing data items: This setting shows items with missing data. Showing items with missing data replaces most INNER joins with LEFT joins. This joins more data than required and can be slower for large databases, but items in the data chain (Samples, Aliquots, Grains, Spots, U-Pb Analyses) missing a sub item (e. g.,a sample with no U-Pb analyses) are still visible. Not showing items with missing data uses INNER joins which are faster for large databases, but items in the data chain (Samples, Aliquots, Grains, Spots, U-Pb Analyses) missing a sub item will not be visible in the table displays, only in drop down lists.

Table Views: A dropdown selects the table to show in the lower list for editing. Checkboxes enable/disable columns to show. Click and drag on a row to change the order columns are displayed in. Available views to edit are: Samples, Columns, Aliquots, Grains, Spots, U-Pb Analyses, and References, along with Edit View variants. Clicking `Reset` reverts the column display settings to GeoCORK default settings for the selected table only.

Font, Font Size, and Table Font Size: Font applies to all text items in GeoCORK. Font size modifies all non-table text. Table font size modifies the size of text inside table cells.

Combobox Height/Width Scalers: These values shouldn't need to be modified. However in unique or odd

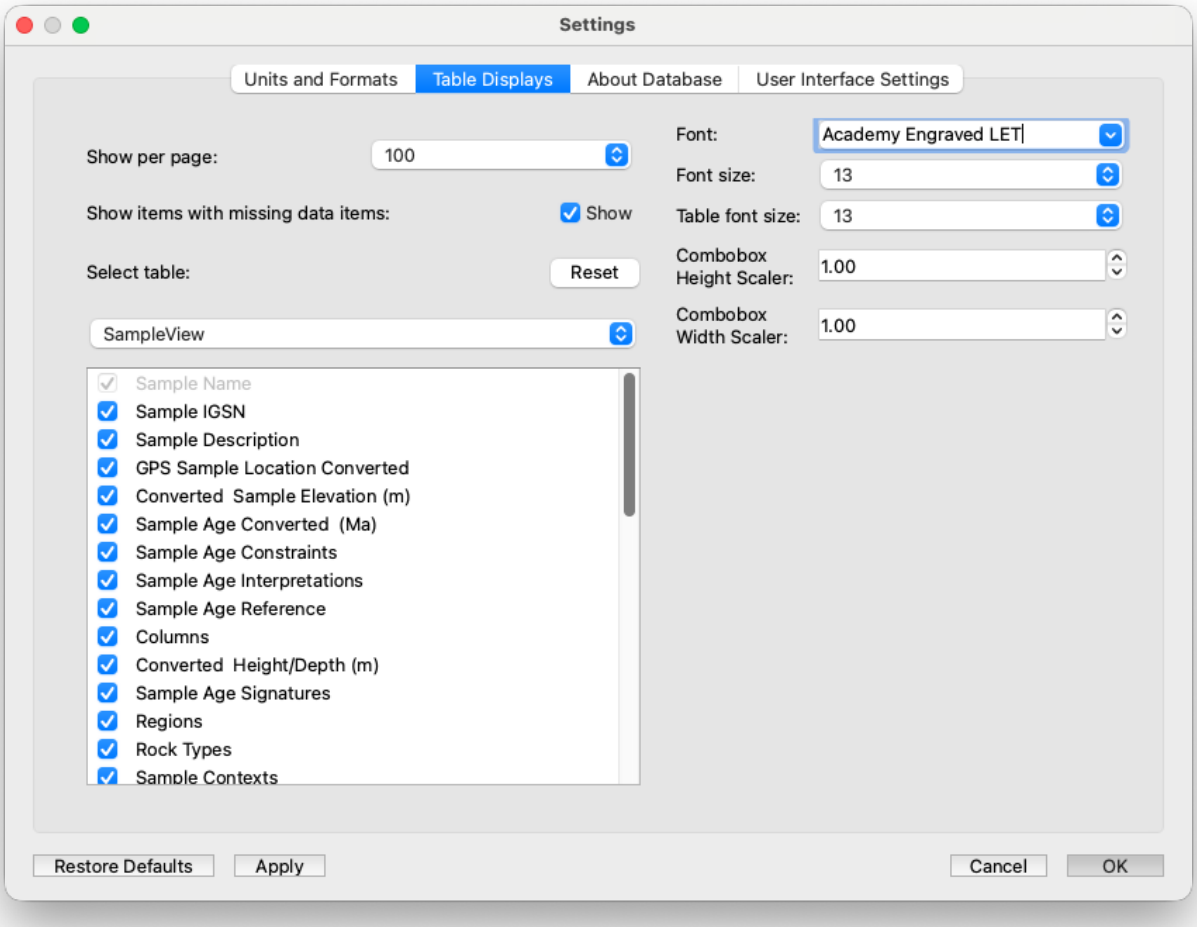


Figure 3: Table displays tab in the settings window.

setups with different screen sizes the dropdowns may display as cutoff or very small. These scalers increase (> 1.0)

2.2.3 About Database

See above about creating a new database. At the bottom of the window, there are two buttons to navigate to GeoCORK support folders on your computer. `Open Logs Folder` will open a File Explorer or Finder window with a log file for each session of GeoCORK. The most recent file is for the current active session of GeoCORK. If you encounter any issues, providing log files can help the GeoCORK team diagnose and fix the problem. `Open Backups Folder` will open a File Explorer or Finder window with a folder for each database file name. Inside each folder are time-stamped backups of that file. Backup your database files regularly (`File > Create Backup`) and restore them directly in GeoCORK (`File > Restore Backup`) or from the backups folder.

2.2.4 User Interface Settings

Tooltips are the hints that display when hovering over an item. If these hints are distracting, you can disable them here.

3 Importing Data

Importing U-Pb geochronology data can be accessed using the main toolbar (`File > Import`) or with `Cmd+I`. This will prompt a new window for data importing.

3.1 Input Requirements

When importing data into GeoCORK there are some specific pre-requirements that need to be done before importing. Listed below are some requirements that should be noted:

- All imported files must only be `.xlsx`.
- One field per column—split combined values such as age±error or GPS DMS.
- Names must be unique across the database.
- All analyses must have a column for either analysis name, spot name, or grain name and a column for either aliquot name or sample name.
- Hierarchical parent-child data are currently all imported as top-level items. Organize parent-child relationships after import.
- Only one non-hierarchical item can be imported per operation. If there is a sheet of information such as maximum depositional age with age information for different methods, first import the best age that will be used as the default age. After importing those ages, change the mapping to import another set of ages for the same sample.

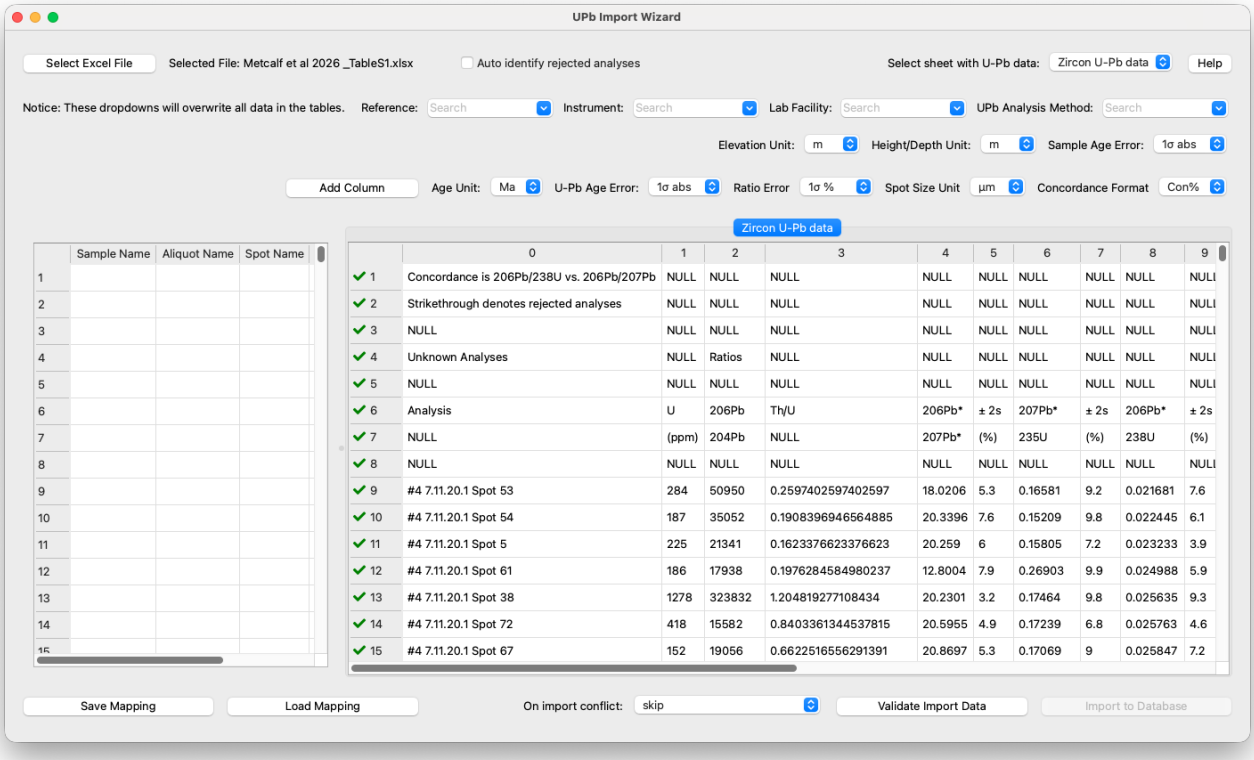


Figure 4: Importer wizard with zircon U-Pb from Metcalf et al. (2026).

3.2 Selecting the File

Choose the import `.xlsx` file. This Excel file can contain multiple sheets, where Sample metadata and U-Pb data are in separate sheets. A prompt will ask which sheet contains the U–Pb data. Large spreadsheets will load more slowly, and we recommend separating into more manageable chunks; empty cells appear as `NULL`.

3.3 Associating Importer Metadata

Double-click any column header to open a column mapper window. The mapper window contains tabs for various metadata and is separated by categories:

- Sample Info
- GPS Info
- Column Info
- Aliquot/Spot/Grain Info
- Reference
- U-Pb Data

Double-click column headers to map them to database fields; a popup window will appear with tabs for different field types, and each dropdown includes auto-complete to help locate values in long lists. Headers that turn green or blue with field names indicate mapped columns, while gray headers with numbers remain unmapped. Duplicate columns may be defined for hierarchical parent-child data, though all entries will be imported as top-level items. To modify a mapping, double-click the column header and choose a different field; to unmap it, select "None."

The top four dropdowns (References, Instruments, Lab Facilities, and U-Pb Analysis Methods) apply globally to all rows in the U-Pb sheet, allowing only one value per field. Values in the dropdown can be changed by right-clicking on the dropdown menu or the combo box. Selecting a value adds an extra column with the data selected in the dropdown and updates the header with the field name (while remaining gray). To remove one of these columns, uncheck it. The two lower rows of dropdowns represent fixed database values such as units and data formats; these are auto-populated with current settings but can be adjusted to match the imported data. While the lower rows of dropdowns do not create additional columns in the sheets, the selected values are saved with the column mapping.

Below are the various color codes for column headers. Columns can be re-mapped or unmapped at any time:

Green Headers	Mapped columns
Blue Headers	mapped and saved
Gray Headers	unmapped, will be ignored

3.3.1 Saving and Loading Mappings

Mappings for all sheets + fixed fields are saved as a JSON file.

Save Mapping	Name or overwrite a mapping.
Load Mapping	Reload a saved mapping.

3.4 Editing Data In Importer

Data can be edited directly within the table, though it is generally recommended to perform edits in Excel or another program to maintain a record of changes. Individual cells can be modified by double-clicking, or multiple cells can be updated at once by highlighting them, right-clicking, and selecting "Set selected cells to value..." to apply a single entry across all selected cells. Entire columns can be edited by right-clicking the header and choosing "Set entire column to value..." or "Set entire column to blank...". Rows and columns can also be removed via right-clicking, while new columns can be added using the "Add column" button or by right-clicking and selecting "Insert column before" or "Insert column after". However, adding or removing columns may impact the ability to save and load mappings in future imports.

3.5 U-Pb Data Sheet

When viewing the U-Pb data sheet, an additional table appears on the left displaying columns for Sample Name, Aliquot Name, Spot Name, and U-Pb Analysis Name, with Grain Name included if it is mapped from the right table. When any of the name-related columns are mapped in the right table, their values automatically populate the corresponding fields in the left table. At minimum, either Sample Name or Aliquot Name and either Grain Name, Spot Name, or U-Pb Analysis Name must be completed. Other name fields can be assigned default values based on the present fields. Each U-Pb analysis is restricted to a single spot, each spot to a single aliquot, and each aliquot to a single sample, enforcing a strict hierarchical relationship.

3.6 Disabled, Rejected, and Accepted Rows

Row state can be changed by selecting rows and right-clicking. Rows are accepted by default, indicated by a green checkmark, and will be imported into the database. Rejected rows apply only to the U-Pb data sheet and can also be marked by double-clicking the checkmark to change it to a red circle with a line through it. Double-clicking again reverts to the accepted checkmark. Rejected rows are still imported but flagged as "Rejected" in the database. Disabled rows, in contrast, are not imported at all; rows to be omitted can either be deleted entirely or simply disabled while remaining visible.

3.7 Validate Import Data

After mapping the data, an initial validation check must be run to identify major conflicts and formatting issues; this process checks for empty cells in the left table for U-Pb data, highlights them in red, and prompts the user to apply default values, which are then filled in and highlighted in purple

for review. It also detects duplicates with conflicting parent relationships in the left table and marks them in red, flags potential duplicates of existing samples, aliquots, grains, spots, and U-Pb analyses in the database in yellow, and verifies that all required fields are properly mapped—particularly ensuring that related data fields are not mapped without their corresponding key identifiers (e.g., sample GPS data without a sample name). Red fields must be addressed before proceeding. Once the data successfully pass this validation, the import button becomes active.

3.8 Conflict Resolution Mode

The dropdown in the bottom center allows the user to control how conflicts with existing database items are handled during import: selecting "Skip" ignores the incoming item and retains the existing data, "Add to" appends new data without overwriting any existing values (for example, adding column information to a sample that previously lacked it while preserving other fields), and "Overwrite" removes the existing data and replaces it entirely with the imported data when a conflict occurs.

3.9 Import

Before importing, the validation check is run again to ensure no changes since the previous check have invalidated the data; if it passes, GeoCORK proceeds with the import. If any formatting issues or unresolvable conflicts are detected during the process, the relevant cells are highlighted, an error message appears, and the import is canceled so the user can correct the problems. The user may also cancel the import at any time by clicking "Cancel" in the progress windows, and once the import completes successfully, the import wizard closes and returns the user to the main window.

4 Viewing and Editing Metadata

4.1 Viewing Tables and Trees

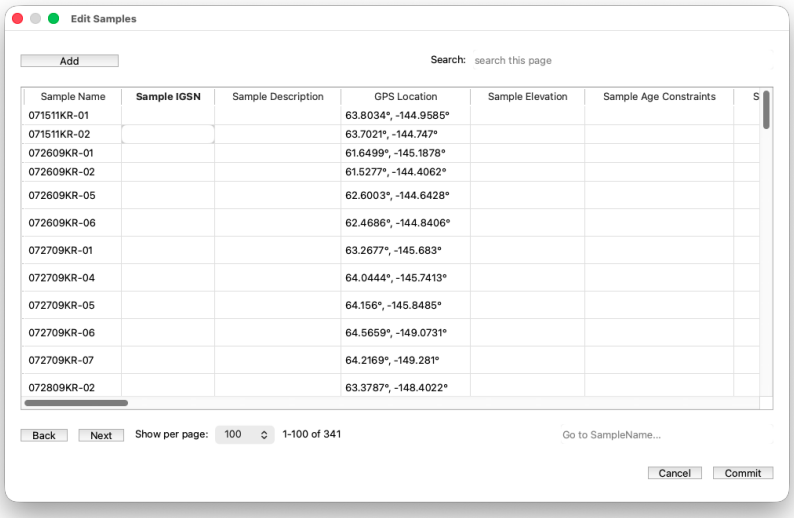
GeoCORK opens to the main window with the Samples table displayed by default, and users can switch between tables using the dropdown in the upper left. Tables are paginated to improve performance, with the number of rows per page adjustable via a dropdown in the lower right. However, hierarchical tables are not paginated and may take longer to load in large databases. The search bar in the upper left allows users to search across any field within the current page, while the lower-left search bar is dedicated to item names and will navigate directly to the corresponding page and row. Clicking a column header sorts items on the current page only.

Data are displayed according to selected units and formatting settings, which can be adjusted in the "Units and Formats" tab of Settings/Preferences, while column visibility and order for various tables (including Samples, Aliquots, Grains, Spots, UPbAnalyses, Columns, and References) can be customized in the "Table Displays" tab. To maintain database integrity, data cannot be edited directly in the viewer. Instead, clicking the edit button opens a separate editing window. The Samples table includes an `Edit Selected` option that opens the Sample Information window, which can

also be accessed by right-clicking a sample and selecting `Edit`, and the `Edit Samples` button enables table-style editing.

Right-clicking within tables provides options to add or edit entries, and in hierarchical views, users can expand or collapse levels as needed. For Samples, additional options allow viewing related aliquots, grains, spots, and analyses, which open in separate tabs and can also be edited. Right-clicking on an aliquot, grain, or spot allows navigation to associated dependent items. The refresh button in the upper left reloads the table to reflect any recent database changes.

4.2 Editing Tables



The edit window is similar to the view window, but may have different columns displayed as defined in the settings. The `Add` button in the upper left opens a dialog to add a new item. Double-clicking text fields like name and description brings up an editable text box. Double-clicking many-to-many tag fields such as contexts brings up a checkable dropdown menu. The dropdown list or tree can be edited by right-clicking. Double-clicking some fields like GPS values opens a dedicated dialog to ensure data integrity across multiple GPS fields.

Multiple values in a column can be changed at once by highlighting the cells in the column, right-clicking, and selecting "Set selected values..." which brings up a dialog that will apply to all selected cells. In a dropdown checklist, any fields that are partially checked apply to only some of the selected items. Only fields that are completely checked or unchecked will apply to all selected items. Cells can also be cleared by right-clicking. Any fields that must be unique like name cannot be bulk edited unless there is common text between them, then only the common text is editable.

Selected items can be merged into one by right-clicking. Select the item to

keep, and all links to the items to be removed are transferred to the item to keep. For example, merging two samples will transfer all associated aliquots, grains, spots, analyses, and sample metadata tags to the selected Sample to keep. However, individual metadata in the edited table (e. g.,Description, GPS, isotopic data) will be lost for items to be removed.

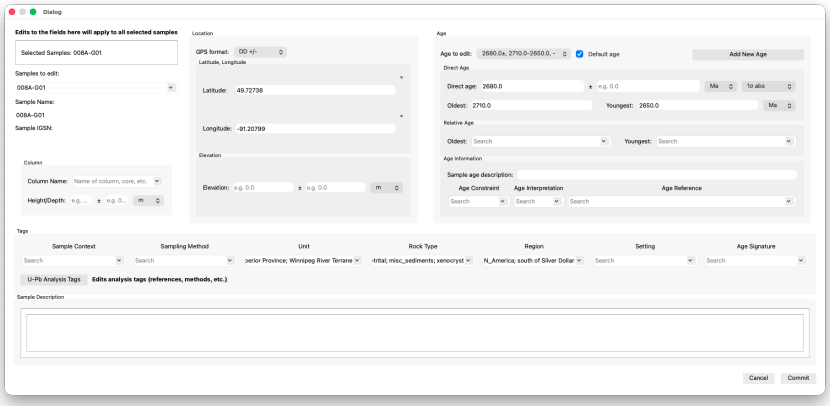
Selected items can be deleted by right-clicking. If selected samples, aliquots, grains, or spots have associated aliquots, grains, spots, or analyses, the user must decide whether to delete these associated items or not. If a selected metadata item (e. g.,lab facility) is associated with other items (e. g.,U-Pb analyses), GeoCORK warns that the item to be deleted is associated with existing items and the association will be deleted as well.

A savepoint is created when the edit window opens. Clicking the window X or the `Close` button confirms discarding changes and then return to the savepoint before the window opened. Clicking `Commit` confirms committing changes to the database, releases the savepoint, and closes the window. If another edit or add window is opened, another savepoint is created. Changes to the database are not final until all edit and add windows have been committed. Discarding changes in the first edit window will also discard all changes made in other edit and add windows.

4.3 Editing Trees

Double-clicking text fields such as name and description brings up an editable text box. Tree items can be moved by selecting one or more items, and then dragging and dropping them in the desired position. Right-click to add an item as a parent, child, before a sibling, after a sibling, or at the end.

5 Sample Information



The `Sample Information` window is a specialty window to modify metadata for one or more samples. To access this window, navigate to the main window and ensure that `Samples` is selected in the main dropdown. Select one or multiple samples by highlighting them in the main view, then select the `Edit Selected` button at the top or right-click and select `Edit`. This

will bring up the `Sample Information` window for the selected Sample(s). If multiple samples are selected, fields with different values will are denoted with `-`.

Selected Samples — All samples selected for editing appear in a dropdown checklist in the upper left, where the active selections determine which records are modified in the Sample Information window. Users can right-click to select, deselect, or delete samples. When multiple samples are selected, unique fields such as sample name and IGSN are disabled, and differing values are indicated with dashes; editing these fields overwrites values across all selected samples. Many dropdown fields can be customized via right-click.

Column Information — Samples may be linked to stratigraphic columns, cores, wells, or similar structures, with new columns added through the dropdown menu. Height or depth values can be entered as positive or negative but must always include associated units.

GPS — GPS coordinates can be entered in multiple formats, and the visible fields change based on the selected format. The system validates entries when the user leaves the group of fields, and any elevation data must include specified units. All available GPS formats include: DD +/-, DD NSEW, DDM +/-, DDM NSEW, DMS +/-, DMS NSEW, and standard UTM.

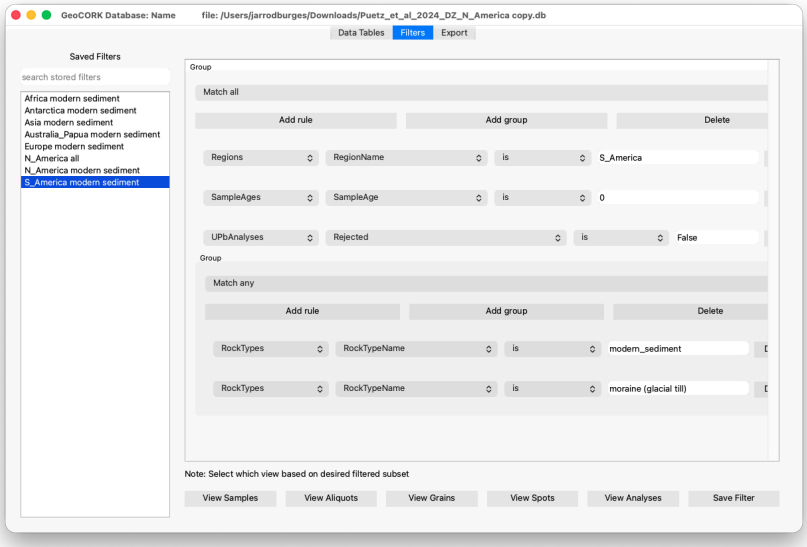
Sample Ages — Multiple ages can be assigned to a sample, and edits applied to multiple selected samples will assign the same age to all, with a warning message provided first. Users can add ages with combinations of direct age values (with error optional), direct age ranges, or relative age ranges, all requiring appropriate units where applicable. Existing age records may be reused across samples, and additional contextual fields allow specification of age constraints, interpretations, and references.

Tags — Several types of tags can be edited from dropdown menus. Multiple tags can be checked or partially checked (line rather than full check). U-Pb analysis tags will open in another window to edit analysis-level information and will apply to all analyses in the sample.

Sample Description — This field is used to store general notes or descriptive information about the samples in a text box.

6 Filtering Data

Filters can be created, edited, and saved within the Filter tab by building queries composed of rules or nested groups of rules, with no limit on complexity. Each group can be set to match all, any, or none of its conditions, allowing flexible query logic. Saved filters retain the query structure rather than the resulting dataset, ensuring that the results automatically reflect any database updates. Filters can be reused by double clicking them from the saved list, and results can be viewed across different data levels, including samples, aliquots, grains, spots, or analyses.



6.1 Building A Query

The `Filter Rule` is a single criterion to query the data by. First select the table then the attribute name within that table. Each type of attribute has different qualifiers to query by and are listed below.

Numerical Rules

Numerical rules are used for numerical data (e. g.,U-Pb data, direct ages, isotope data, concentrations, distances/sizes, dates). For ranges, enter two values separated by a comma (i. e.,1.0,2.0 or 2025-01-01,2025-12-31)). The order of the range does not matter.

- is equal to
- is not equal to
- is greater than
- is less than
- is between
- is not between
- is blank
- is not blank

Categorical Rules

Categorical rules are used for text fields (e. g.,Units, Rock Types, Regions).

- is equal to
- is not equal to
- starts with
- ends with
- contains

- does not contain
- is blank
- is not blank

Calendar Rules — Calendar rules are used for the `Created` and `Modified` attributes found in every table. These are useful for quickly finding and editing recently imported or modified data. For ranges, enter two values separated by a comma (i. e.,1.0,2.0 or 2025-01-01,2025-12-31)). The order of the range does not matter.

- is on
- is not on
- is after than
- is before than
- is between
- is not between

Grouped Rules — Within the `Filters` tab `Filter Rules` can be grouped together under a single `Grouped Rule`, and allows complex nested filtering. The `Grouped Rule` has three options for how data returned from each rule is handled.

Match All — The Match All option returns the intersection of all the data from the rule widgets.

Match Any — The `Match Any` option returns the union of all the data from the rule widgets.

Match None — The `Match None` option returns the inverse intersection of all the data from the rule widgets.

6.2 Saving a Filter

Once a filter is in the `Filter` tab's query builder, it can be saved by clicking the `Save Filter` button in the lower right. In the popup window, enter the name and description. The name will be displayed in the left column of the `Filter` tab, and the description will display when hovering over the name. Filter names must be unique. Load a saved filter by double clicking the name in the left column. If a filter is loaded and `Save Filter` is clicked, GeoCORK will ask if you want to update the current filter or create a new filter. This allows for editing an existing filter or modifying an existing filter to create a new one. Filters are shown in alphabetical order in the left column and can be deleted by right clicking.

6.3 Viewing a Filter

Once a filter is in the `Filter` tab's query builder, it can be viewed. Since GeoCORK is hierarchical in how Sample to U-Pb data is stored there are options to view the filter at any given level.
This also means that if filters are set at the U-Pb data level, but Samples are viewed, there may be samples returned that are included because a single U-Pb analysis matches the criteria. This behavior is intended and the reason we include multiple options of hierarchy viewing.

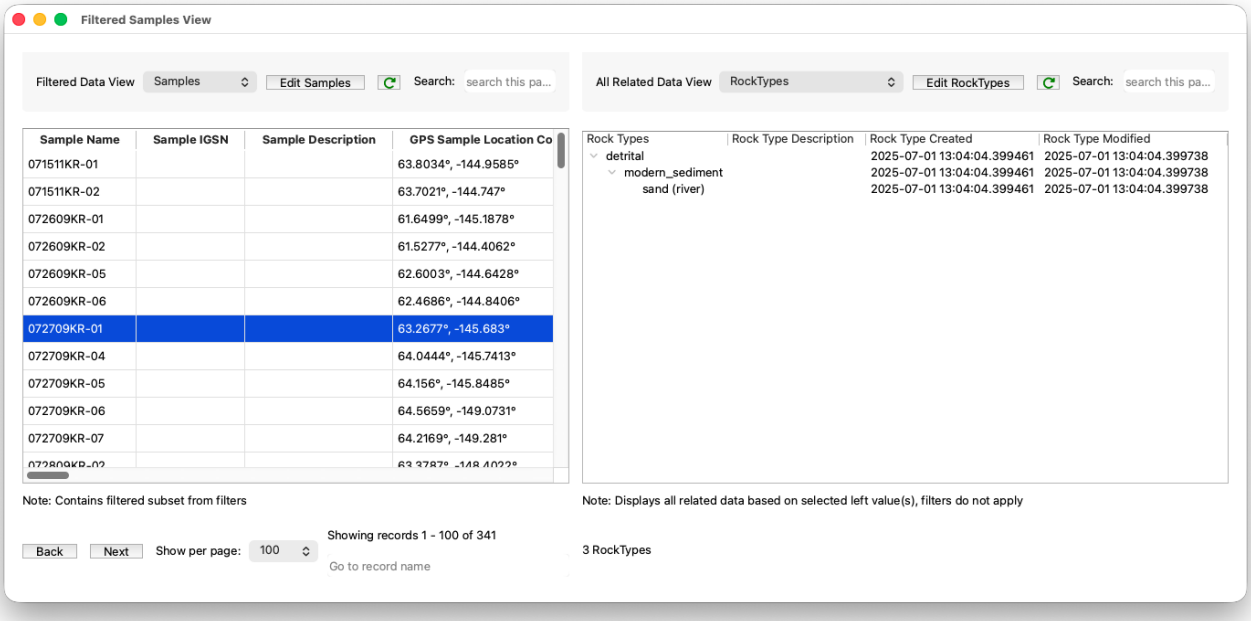


Figure 5: Dataviewer widget.

The window after pressing one of the `View ...` buttons is called the `Data Viewer`. These windows are useful for editing multiple items at once or viewing filter results before export.

View Samples — Viewing samples will return any samples that match the given criteria or that are linked to aliquots, grains, spots, or analyses that match the given criteria.

View Aliquots — Viewing aliquots will return any aliquots that match the given criteria or that are linked to samples, grains, spots, or analyses that match the given criteria.

View Spots — Viewing spots will return any spots that match the given criteria or that are linked to samples, aliquots, grains, or analyses that match the given criteria.

View U-Pb Analyses — Viewing U-Pb analyses will return any U-Pb analyses that match the given criteria or that are linked to samples, aliquots, grains, or spots that match the given criteria.

Data Viewer — The `Data Viewer` is split into left and right tables. The left table contains the queried/filtered data and can be switched to the desired hierarchy level by using the top drop down. The right table has a top drop down to switch between various metadata tables (e. g.,Rock Types, Units, Spot Contexts, etc). This table is dynamic and will change depending on the selected rows in the left table. Using the combined left and right tables allows easier understanding of what is associated with a given data point quickly. For instance, if Rock Types is selected for the right table, selecting the various left data points (either multiple or single selection) will dynamically change which Rock Types are shown, indicating association.

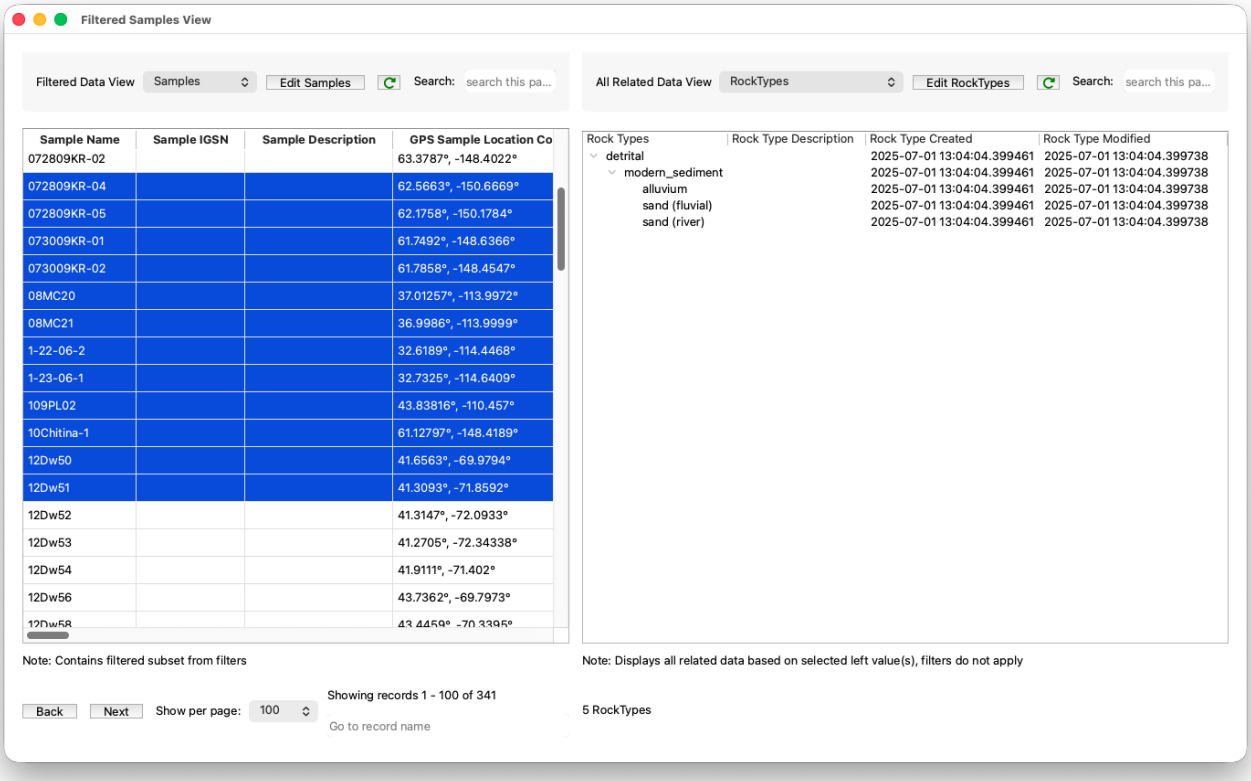


Figure 6: Dataviewer widget with selected values to show dynamic right table updating.

Editing the left table will bring up an edit window with all items that pass the given criteria, making it easy to filter and bulk edit. Editing the right table will bring up an edit window with all items in the table, whether or not they are associated with filtered items in the left table. Multiple `Data Viewer` windows can be open at once to compare the results of different filters, but refreshing any of them will update the window based on the current filter in the `Filter` tab, regardless of the filter selected when the window was opened.

7 Exporting Data

Exporting data from GeoCORK into an Excel or comma-separated value (CSV) file is done through the `Export` tab in the main window. The exporter includes many pre-loaded formats for popular data visualization tools:

- detritalPy
- IsoplotR
- DZstats, DZmix, DZmds, DZnmf
- AgeCalcML Concordia

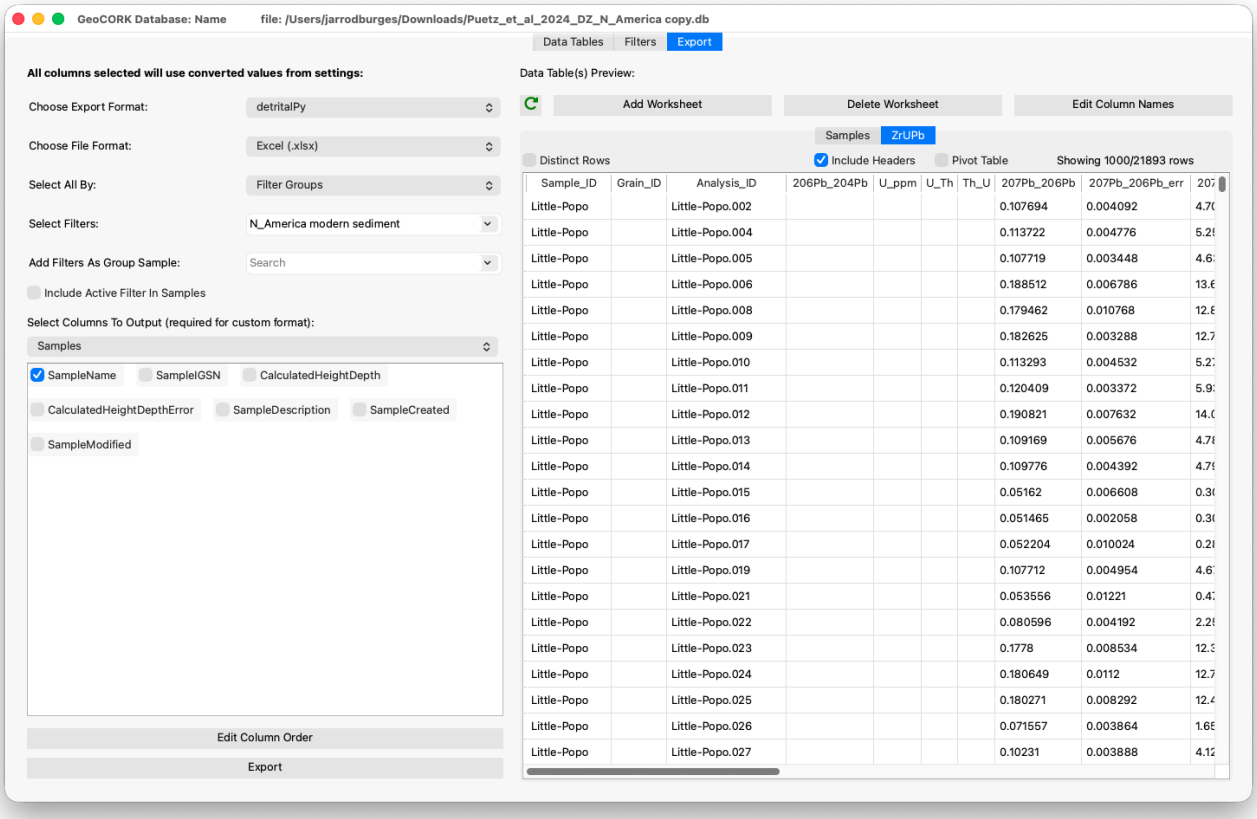


Figure 7: Exporter tab with detritalPy selected and ZrUPb sheet selected.

- GeoCORK database format

For the formats above, the age errors must be in **absolute**, not in percent (e. g.,1σ and Ma %). If the current settings do not match the settings required for the selected format, GeoCORK will ask to update the settings to match the expected export format. The current unit and format settings are shown in the lower left.

7.0.1 Selecting Samples for Export

Selecting data for export can be done through individual sample selection, selection by saved filters, selecting by samples and saved filters, and through active filters.

Selection by Samples — The simplest method is to individually select samples through the `Select Samples` drop down menu. Check boxes within the drop down to select each sample. Right-clicking within this drop down will show options to select or deselect all samples.

Selection by Samples with Filtered U-Pb Data — Samples selected by the above means can be further filtered using the `Select Additional Filters` drop down. This means selecting various Samples and applying additional saved filters (e. g.,only accepted analyses, concordance limits, U/Th ratios, contexts).

Selection by Filters — Selecting the `Select all by` drop down and changing to `Filter Groups` allows the selection of samples by filters. Check boxes within the drop down to select each filter. If multiple Filters are selected, the result is a union of sets. All data matching any of the selected filters will be selected. This means that Filters A, B, and C resulting in U-Pb data subsets `X, Y, and Z` will be joined such that:

$$\textit{U-Pb Data} = X \cup Y \cup Z$$

Add Filters as Group Sample — Check boxes within the drop down to select each filter. Each selected filter is returned as a single "sample" with the name of the filter. This enables exporting a compilation of data as one item and can be combined with selected samples or filters.

Active Filter — When the `Active Filter` checkbox is selected, all other filter criteria are overridden. The active filter pulls the filters from the `Filter` tab and applies them to the `Export` tab. This is best used to quickly filter, export, and plot data using multiple combinations.

7.1 Export Preview

Once a format and data are selected for export, the right table populates with a preview of the first 1000 lines of data to export. If the selected export format is `Database`, the right table is replaced by the `Data Viewer` with side-by-side tables for items and attributes.

7.1.1 Custom Export Format

Custom export formats can be created for maximum flexibility. Select `Custom` from the `Choose Export Format` dropdown.

File Format — Select the output file format (.xlsx or .csv) from the `Choose File Format`

dropdown.

Fields — Below sample and filter selection in the left column, select a table from the drop down and check fields to display. Fields are added as columns in the order selected, but the order and names of columns can be changed as explained in `Export Display` below.

Some fields have `Display`, `Calculated`, or `Converted` variations. The `GPS Locations` table illustrates these with the following fields:

- `GPSLocationConverted` - The combined GPS latitude and longitude fields with directions and converted to the current GPS format selected in the settings.
- `GPSLocationDisplay` - The combined GPS latitude and longitude fields with directions in the GPS format selected when entering data.
- `CalculatedLatDisplay` - The latitude fields combined with direction and converted to the current GPS format selected in the settings.

The U-Pb analysis table also has the following fields:

- `CalculatedBestAge` - The best age for the analysis converted to the current age unit selected in the settings.
- `CalculatedBestAgeFilled` - The best age for the analysis or the age selected to fill empty best age values according to the settings, converted to the current age unit selected in the settings.

Worksheets — The exporter supports multiple worksheets. Click `Add Worksheet` to create a new worksheet to the left of current sheets. Double click a sheet tab name to edit the worksheet name. Selected samples and filters apply to all sheets, but each sheet can display different fields.

Export display — Click `Edit Column Order` below the field check boxes on the left to reorder columns. Click `Edit Column Names` in the upper right to change the display names of columns. Below the worksheet tab names, there are check boxes for `Distinct Rows`, `Include Header`, and `Pivot Table`. By default, each row represents an analysis. However, if only sample fields are selected, there may be many duplicated rows for each analysis from the same sample. Selecting `Distinct Rows` collapses duplicates so that only rows with distinct values are displayed. `Include Header` specifies if the header names should be included as the first row in the export file. Some export formats (DZ suite) expect repeating groups of columns for each sample (e. g.,Sample 1 Age, Sample 2 Age Error, Sample 2 Age, Sample 2 Age Error). GeoCORK accomplishes this with the `Pivot Table` option. The first column field should be the sample (or grouped filter) name, and all other field columns will be repeated with each sample name as the prefix.

7.2 Exporting

When the data are ready for export, click the `Export` button in the lower left. Choose a location and file name for the exported file. If the export file is .xlsx or .csv, the file will automatically open in the computer's default application for inspection. If the export format is `Database`, a File Explorer or Finder window will automatically open to show where the exported file is

saved. To view the exported file, close the current Main Window to return to the Landing Page and select the new .db file to open.

8 Troubleshooting & Common Issues

8.1 Understanding Navigation and Names

Hovering over buttons, table headers, and items in dropdown menus often displays a tooltip, popup text to give more information. For database items, the description field is used as the tooltip. If you notice places where additional tooltips would be helpful, let us know atCo the GeoCORK GitHub.

Video tutorials will be coming in the future, and on-screen tutorials are planned for future releases.

8.2 Issues with Opening Views

As GeoCORK expands in features or backend changes some settings may break for some researchers. While we make our greatest attempts to minimize this, sometimes settings may need to be reset. If there are weird displaying tables, column selection, or database views causing errors we recommend resetting to default settings. This should fix many problems that sometimes give the most obscure issues.

8.3 Importing data

If data was imported incorrectly or by mistake there are a few ways to go about it.

If data was imported incorrectly or some columns were missed, as long as the SampleName was imported correctly it is easily fixable. Try importing the whole file again using the `Overwrite` conflict resolution mode or importing only the missing data using the `Add to` conflict resolution mode.

If data were imported by mistake, we recommend sorting samples by newest created in the main window and deleting them. This deletion will recursively remove all aliquots, grains, spots, and U-Pb analyses associated with a given sample. To remove other metadata imported, follow similar methods in the other tables. Note that trees are not sortable, but new items will be added as top-level items at the end.

8.4 Opening and Reading Log Files

In the settings dialog under the `About Database` tab there are two buttons at the bottom: `Open Logs Folder` and `Open Backups Folder`. These will open a file explorer window and navigate to the respective folders.
The logs folder will create a new log each time GeoCORK is launched. Opening the log files will provide detailed context on the inner workings of GeoCORK, including more detailed errors that are useful for debugging. If GeoCORK is open, the most recent log file is the active session.

8.5 Reporting Bugs/Features/Ideas/Suggestions to GitHub

We welcome all feedback to our GitHub repository at GeoCORK GitHub. Please submit detailed issues with the bug, feature, idea, or suggestion. For issues, please include all diagnostic information including the log file at the timestamp that had the issue.